

# Nandana Dileep

AI Engineer

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## Experience

### AI Engineer, JLR Technology & Business Services India

July 2024 - Present

- Core developer on an eight-person team building Sarvam, an AI platform for automotive requirements, system design, tests, and compliance; adopted by major vehicle programs and presented to JLR's CEO.
- Built Octopus to prioritize test cases using past execution results and relevance to code changes; **adopted across JLR**.
- Deployed an ensemble classifier to identify underused test rigs from 500K+ daily telemetry events, enabling asset consolidation and an **18% cost reduction**.
- Built a C++ GNSS processing pipeline using GTSAM for vehicle position estimation; presented the work at the JLR Tech Symposium.

### Software Development Engineer Intern, Amazon

May 2023 - July 2023

- Designed and deployed a latency analysis service for Alexa using AWS Lambda, Coral, and OpenSearch.
- Built an end-to-end testing framework using JUnit 5 and Mockito.

## Selected Projects

### Mycelium - Reliable Tool Execution for AI Agents

[GitHub](#) | [PyPI](#)

- Maintain an open-source Python runtime for agent tool execution, with guards against duplicate actions, unsafe retries, stale context, and invalid tool calls.
- Supports LangGraph, CrewAI, and custom Python agents; **30k downloads on PyPI**.

### Meridian - AI Hiring Platform

JLR Global Techathon Winner, 2026

- Built a platform that compares job descriptions with resumes, maps skill gaps for recruiter review, and delivers adaptive technical assessments.

### LLM Serving Engine on Apple Silicon

[GitHub](#)

- Built a local LLM serving engine with MLX, continuous batching, and PagedAttention-inspired KV cache management.
- Implemented OpenAI-compatible chat completions and published benchmarks against vLLM and llama.cpp.

### KAIROS / Auto-Agent - Background Coding Agent

[GitHub](#)

- Built an agent that monitors software projects, identifies and executes tasks, and maintains project memory across runs.

### Self-Learning Content Writer

Personal project

- Built a writing assistant that adapted to each user's voice by learning from writing samples, revision history, and editorial feedback.
- Deployed and maintained the application; set up the database with pgvector for semantic embeddings and style profiles.

## Technical Skills

**Languages:** Python, C++, TypeScript, SQL

**AI / ML:** PyTorch, Transformers, LangGraph, CrewAI, LlamaIndex, RAG, MCP, MLX, scikit-learn

**Backend / Data:** FastAPI, Flask, PostgreSQL, Redis, Neo4j, OpenSearch, FAISS

**Engineering:** AWS, Docker, Git, Linux, Pytest, React, Next.js

## Education

Indian Institute of Technology Madras

Aug 2020 - May 2024

Bachelor of Technology, Civil Engineering

## Selected Writing

[Mycelium: Turning AI Agents into Reliable Systems](#) | [LLM Serving with MLX](#)